L7.4: Inner products and norms on a vector space

Transcript Language

English

Hello, and welcome to the Maths 2 component of the online course on data science.

In this video, we are going to talk about inner products and norms on a vector space.

This is linked to our previous video which was on length, angles and dot product. So, we will see

that the notion of an inner product generalizes what is a dot product and the notion of a norm

generalizes what is the length. So, the point here is that we c[illegible]p this for any vector space.

So, let us start by the, by defining what is an inner product. So, an inner product on a vector

space V is a function, which is, which takes two vectors as input, so that is why the dot

comma dot, so in place of the dots, you will have vectors. So, it is a function from V cross V to R.

So, it produces a real number. For each pair of vectors, it produces a real number,

and it satisfies the following. So, if you take into, if you apply this to v comma v,

that means you take just one vector, and you think of it as the tuple v comma v,

then the inner product of v with itself, that is how we will say v comma v this angle brackets,

this must be positive, strictly greater than 0

if v is non-zero, and it will be 0, precisely when v is equal to 0.

And then you have these three things, which are, which I have put together because they are called,

this is saying that it is by linear. So, these three things say it is by linear.

So, v1 plus v2 is comma v3 is v1 comma v3 plus v2 comma v3.

And v1 comma v2 is v2 comma v1. And c times v1 comma v2 is c times v1 comma v2, which

because of your previous thing can be written also as v1 comma c times v2, where c here is a

real number, it is a scalar.

So, I will also point out that using these two, we can get that v1 comma v2 plus v3

is equal to v1 comma v2 plus v1 comma v3, because v1 comma v2 plus v3 is v2 plus v3 comma v1,

which is then v2 comma v1 plus v3 comma v1, and then you can flip those again to the other side.

So, now, these three together

are what is called bilinear. So, this is a bilinear map, the first and third.

So, a vector space together with an inner product is called an inner product space. So,

I will underline this. This is called an inner product space. This is a definition.

And I maybe I will just put in the conclusions we do. So, the conclusions we do before where that,

so first of all here, this is not a conclusion, this is a hypothesis, c is a scalar. And the conclusion out of this was that v1 comma v2 plus v3 is v1 comma v2 plus v1 comma v3.

So, the dot product is an example of an inner product. This is what I started with. So,

recall that the dot product of u and v in Rn is u dot v is u1v1 plus u2v2 plus unvn.

I apologize for this ugly looking full stop which should have been here. And this is a function

which is given by u comma v is u dot v, angle brackets u comma v is u dot v.

So, why is this inner product? So, we have to verify the three axioms,

the four axioms we had. So, we know already that if u comma u is, so this is u1

square plus u2 square plus un square, for real numbers squares must be positive, sum of squares

is positive. So, that means this is greater than 0 if u is not 0 and this is equal to 0

exactly when each ui is 0. So, ui is 0 for all i, which happens precisely when u is 0.

So, we approved the first one. And the other three just follow from the form of this equation.

So, if you have u plus u prime dot v, you can check that this is actually u dot v

plus u prime dot v. And similarly, if you have u dot v, this is same as v dot u that is clear from

the equation, because real numbers you can, they commute under multiplication. And then

clearly if you have cu dot v, this is c times u dot v because c multiplies into each component.

So, for each component you will have it over there and then you can take it out. So, this completes

the proof that this is indeed a inner product. So, the dot product is an inner product.

So, let us now look at an example of an inner product on R2, which is, this is not the standard

inner product that we saw, the dot product that we saw earlier. This is slightly different

example. So, here is the definition, u comma v is x1y1 minus x1y2 minus x2y1 plus 2 times x2y2,

so where u is x1. This is a typo here. So, u here is x1 comma x2 I think and v is y1 comma y2.

So, the standard inner product that we saw was, so if you do u dot v, u dot v is x1y1 plus x2y2. So, sorry, in terms of, is x1y1 plus x2y2. The inner product here

is slightly different as you can see. So, why is this an inner product, how do

I prove that? So, this is proved by looking at the following. If you look at this expression,

you can write down a matrix for this expression. So, you have x1x2 and then you have y1y2.

And then let us see. So, you have x1y1 the coefficient is 1, x1y2, the coefficient is

minus 1, I believe, this should be minus 1 here. And then this is minus 1 and then this is 2. So,

if you multiply this, let us see what we get. You get exactly,

so the first term is x1 minus x2, the second term is minus x1 plus 2 times x2 times y1y2

and if you work out what this is, this is x1y1 minus x2y1 minus y2x1 plus 2 times x2y2. So,

this is exactly what we have up here. This expression and this expression are the same.

So, this is a different expression from the inner product, which is x1y1 plus x2y2. And now how do I

use this matrix form to conclude that it is an inner product. So, the reason, so I can use a matrix form to calculate that this is an inner product because the bilinearity part follows

directly. And the symmetry part follows because this is a symmetric matrix. So, if I want to do

y1y2 comma x1x2, then you can see that this is exactly writing y1y2 here, writing the same

matrix here and x1x2, but this is a real number. So, I can, so this is the same as its transpose.

This is the same thing transpose. And if you do the transpose, you will get this, because

the matrix in the middle is a symmetric matrix. So, its transpose is itself.

So, that is why it is symmetric. That is the second property. And then bilinearity is because

matrix multiplication respects addition. So, if you have,

so it distributes over addition. So, if you have x1x2, let us say, plus x1 prime x2 prime,

then over here you will, that will distribute and it will give you what you want. That is the

first thing. And then c times x1x2 again because constants come out of matrix multiplication,

it will follow from there. So, I will urge you to check that fact, because these two expressions are the same. So, the key point here is these two expressions here are the same. So,

maybe I will draw that better. These two expressions are the same.

So, you can check it from there. So, the only remaining thing is

that this is always positive. So, if you put x1x2 in place of y1y2 what do you ge?

So, you get x1 square minus x1x2 minus or rather plus x1x2 so that is 2 times x1x2

plus 2 times x2 square, so this is x1 square minus 2 times x1x2

plus 2 times x2 square and you can write this as x1 minus x2 square plus x2 square. So,

this is a sum of square. Again, it must be positive. If it is equal to 0, then both the

terms which are squares are 0, that means x1 minus x2 is 0 and x2 is 0. Once x2 is 0,

x1 must be 0. So, both x1 and x2 are 0. So, that tells us that it is 0 exactly when

u is 0. So, this proves that it is an inner product on R2.

Let us define what is a norm. So, norm on a vector space is a function. It is indicated by that double lines. And we have seen these double lines in a previous, in the previous video,

x goes to norm x, which satisfy the following conditions.

So, norm of x plus y is less than or equal to the norm of x plus norm of y; norm of c times x

is the absolute value of c times norm of x. So, remember, in the previous video this norm

was supposed to represent length. So, since it represents length, it better be positive.

And so if you have, if you scale it, the length better remain positive. That is what is being

reflected here. So, positivity is coming in the third axiom. Norm of x is always greater than 0.

And it is 0 precisely when we are dealing with the 0 vector, if and only if the vector x is 0.

So, let us recall the length of a vector. This is the comment I made in the previous slide

u. It is given by this expression root of x1 square plus x2 square plus xn square.

And this is actually a norm. So, we have to say that the three previous conditions are satisfied.

As I said, if you multiply this by c,

this is clearly root of c square x1 square plus c square x2 square plus c square xn

square. And the c comes out but with an absolute value. And the remaining part is norm of u,

length of u. This should have been two bars. Let us see when this is 0. This is 0,

we have actually checked this, so the length is 0 exactly when u is 0. This is something we know

which leaves us with the triangle inequality. Namely, if you take two vectors, u plus v,

then you compute the sum. So, this is root of

x1 plus y1 square plus x2 plus y2 square plus xn plus yn square.

And on the other hand, we have norm of u plus norm of v, which is x1, root

x1 square up to xn square. This is root y1 squared plus all the way up to yn square.

So, I want to say that these, this sum is less than or equal to that sum.

And this is something that we know is true. And I will maybe not do a proof of this.

So, what you can do is you can square both sides. If you square both sides, you will get x1 plus y1 square plus all the way up to xn plus yn square. This side you get x1 square

plus all the way up to xn square plus y1 square all the way up to yn square

and then plus 2 times this root and then y1 square plus all the way up to yn square.

And then, so to compare these, so I want to compare these. So,

to compare these you can expand the brackets. If you expand the brackets and you cancel off these

square terms. You are going to have to compare 2 times x1y1 plus 2 times x2y2 plus 2 times xn yn

with this side. And now I will leave it at this point to.

So, you can cancel out the 2s as well. So, you can, you have to compare this term

with this term here. And I will may be not get into a proof that this is actually,

this is less than or equal to that. It is a very, something you can do. So, the length function is

a norm on Rn. So,

let us look at a norm on Rn which is different from the length function. Namely, you can take the absolute values of the coordinates. So, norm of u1, so this 1 is supposed to,

I think this 1 has not come out the way I intended it to be. This should have been the subscript,

but it did not come out as far as I wanted it. Anyway, so define norm of u

to be this, the absolute value of x1 plus absolute value of x2 plus absolute value of xn then this is

a norm. So, let us check that this is a norm. So, first of all, I want to check that

if norm u1 is 0, this is the third axiom, what happens. Well, this is a

sum of non-negative integers. So, this happens exactly when each xi is 0 in absolute value,

which is exactly happening when xi is 0 for all i, which is exactly saying that u is equal to 0.

So, if this norm is 0, if this function is 0 on u, then u better be 0. And,

of course, if the, if u is 0, then the function is 0. So, this is one thing that we have checked.

The other is, if you multiply this by a constant c, by a scalar, what happens. Well, each term is multiplied by c. So, I get this expression. Excuse me, I should put in the dots,

this expression. And I can take my c out common, because remember that the

absolute value of two numbers multiplied is the absolute value of each multiplied. So,

you take c out, absolute value of c out common and you get x1 plus x2, all these in absolute value,

so absolute value of x1 plus absolute value of x2 all the way up to n which is

the definition of the norm here. So, c times this function. So, this gives you one more thing.

And then the last one that we have to do is the triangle inequality, which is saying that if you

have u plus v, you take this. Well, let us write that down, that is x1 plus y1 plus x2 plus y2

plus xn plus yn, where xis are the coordinates of u, yis are the coordinates of v.

But we know that this is less than or equal to x1 plus

y1, this is less than or equal to x2 plus y2. And then if you rearrange all these terms,

you get that this is u1 plus v1.

So, this tells you that it is a norm. So, it is a fairly simple check that it is a norm.

So, finally, let us explore the relation between the inner product and

the norm. So, suppose you have v which is an inner product space, so it has an inner product

which we have represented by these angle brackets. Let us define this function.

So, this function is norm of v is equal to square root of the inner product of v with itself

then the claim is that this is a normal v. Let us work this out,

why this is a norm. So, first of all, norm

of v is 0. This happens exactly when the square root of the inner product of v with itself is 0,

which happens precisely when the inner product itself is 0. But remember, it is an inner product.

This happens precisely when v is 0. So, I want to prove that this is always positive if

v is non-zero, but remember that v comma v is, so if v is not equal to 0,

v comma v is greater than 0, that means square root of v comma v is strictly greater than, we are always looking at the positive square root, which means norm of v is strictly. So, the

positive, non-negative and positive part is proved, that axiom is proved.

So, the other two axioms are the triangle inequality.

And if you multiply a scalar, what happens? So, let us look at scalar.

So, for the scalar, so we have c times v, by definition this is cv comma cv.

But as we know c comes out, so this is root of c. So, it comes out of the first

one and it comes out of the second one. So, c times c, excuse me, this is norm of v,

inner product of v comma v. So, this is root of c square, root of v comma v. The second part

is exactly the norm of v and the first part is mod of, meaning the absolute value of c,

because you are taking root of c square. So, if c is a negative number, after taking square root,

you always look at the positive part. So, it is the absolute value of c times norm of v. So,

this is the another one of the axioms. And finally, we have, if you have v plus w, by definition this is v plus w comma v plus w square root and this is,

so you have four terms. You have v comma w plus v comma, w comma v plus w comma w

which if you rewrite this is v comma v and this is remember its inner product. So, it is,

so the first, we got the first thing by bilinearity, and then, because it is symmetric, we

get this is 2 times v comma w plus w comma w. So, let us square this.

So, if you square this, you get v comma v plus 2 times the inner product plus w comma w,

but this is exactly norm of v square plus norm of w square plus 2 times v comma w

and this is less than or equal to norm of v square plus norm of w square plus

2 times norm of v, norm of w. So, this needs a little bit of

work. And then what you will get is that this is norm of v plus norm of w

the whole square. So, we have to prove that norm of v plus w the whole square

is norm of v plus norm of w is less than or equal to norm v plus norm w the whole square.

The only thing missing here is why is, this thing less than or equal to this thing here.

So, that needs a little bit of work which I will skip for now. So, I guess this tells you

the relation between inner products and norms. So, the main point here is that once you have

an inner product, it induces a norm. This is what you have to keep in mind. Thank you.